Summary: Generative AI & Health Misinformation

- **Objective:** To systematically synthesize empirical evidence (January 2023 – August 2025) exploring how generative AI alters the creation, dissemination, and regulation of false health narratives across technical, sociotechnical, and governance layers.
- **Data Base:** 15 empirical studies across 10 major databases (e.g., MEDLINE, Scopus, IEEE Xplore, arXiv) meeting strict PRISMA guidelines.
- **Core Finding:** Generative AI dramatically lowers the barriers to producing highly credible, localized, and mass-scaled health disinformation. Furthermore, it introduces unique **governance loopholes**—particularly at the API and marketplace levels—that render front-end User Interface (UI) guardrails insufficient.

1. Production Economics & Technical Vulnerabilities

The paper divides user interaction into two primary attack surfaces, both of which exhibit significant vulnerabilities to generating harmful content:

Front-End User Interfaces (UIs)

- **Jailbreaking:** Standard consumer-facing chat interfaces remain highly vulnerable to simple jailbreaks that easily bypass safety filters to elicit false health narratives on highly polarizing topics.
- **False Fluency:** AI models generate text that mimics professional expertise, utilizing uncertainty framing, scientific structures, and fabricated citations. This creates an authoritative "veneer of credibility" that easily tricks human surface-level inspection.

Programmatic Access (APIs) & Marketplaces

- **System-Instruction Vulnerabilities:** Programmatic access via APIs represents a severe, unmoderated risk surface. When subjected to malicious system-level commands, top-tier models were successfully converted into functional health disinformation agents, generating wrong or malicious answers in **88% of trials**.
- **Throughput Scalability:** Automated pipelines demonstrate unprecedented campaign speed. In a documented case study, a multi-topic adversarial prompt generated **102 distinct vaccine/vaping blog posts (>17,000 words) in 65 minutes**, alongside **20 realistic images in under 2 minutes**.
- **Marketplace Loopholes:** There is an acute lack of corporate oversight and safety configurations regarding custom AI assistants deployed in public software marketplaces.

2. Platform Propagation Dynamics

The spread of AI-generated misinformation ("AI-misinfo") behaves differently than traditional, human-authored fake news:

- **The Decoupling Paradox:** Controlled user studies revealed that while participants often rated AI-generated COVID-19 fake news as slightly *less* accurate than human fakes, their **sharing intentions remained identical**. Dissemination behavior is driven by emotional resonance, style, and novelty, rather than factuality judgments.
- **The "Many Small Seeds" Strategy:** On platforms like X (formerly Twitter), AI-disinformation typically originates from **smaller, low-follower accounts**. However, due to optimized emotional/entertaining tones that exploit platform recommender algorithms, these posts have a significantly **higher likelihood of going viral** without requiring traditional, high-leverage botnets.

3. Evaluation of Mitigation Strategies

The review evaluates current countermeasures across three distinct operational layers:

Layer	Strategy Evaluated	Efficacy & Legal/Technical Limitations
Technical	Front-end Guardrails & Automated Classifiers	**Low-Moderate.** UI safety updates are short-horizon and easily outpaced by adversarial adaptation. Current detectors fail because AI text lacks traditional "bot" signatures, and state-of-the-art (SOTA) models struggle to track multimodal provenance (text + image).
Sociotechnical	Platform Warning Labels & UX Prompts	**Context-Dependent.** Generic "AI-generated" content labels can reduce perceived accuracy, but blunt implementation risks creating a "chilling effect" that accidentally suppresses legitimate medical data.

Governance	Audits & Provider Transparency	**Nascent/Uneven.** Developer reporting channels remain highly inconsistent, and there are profound transparency gaps regarding model safety finetuning data.

4. Legal & Regulatory Recommendations

To protect health information ecosystems equitably, the synthesis argues that law and policy must pivot away from standard "model card transparency" toward verifiable liability and enforcement mechanisms:

1. **Mandatory Audit Trails & System Logging:** Regulators should mandate that AI providers maintain immutable, post-hoc logs of system prompts, tool utilization, and retrieval sources for any AI applications running in health-adjacent deployment. This facilitates legal attribution and remediation after an incident occurs.
2. **Pre-Deployment Red-Teaming Standards:** Policymakers should establish strict legal compliance frameworks requiring model developers to rigorously stress-test models against system-prompt manipulation and custom marketplace exploits prior to commercial release.
3. **Public Provider Scorecards:** Introduce legally backed accountability scores tracking provider response latencies to reported safety incidents and requiring granular verification of safety data curation.
4. **Addressing Asymmetries (Linguistic & Multimodal Bias):** Current empirical evidence is heavily concentrated in high-income, English-language text environments. Regulators must ensure compliance frameworks protect lower-resource languages and account for audio/video deepfakes, where clinician/doctor voice impersonation presents an unresolved threat to consumer trust.